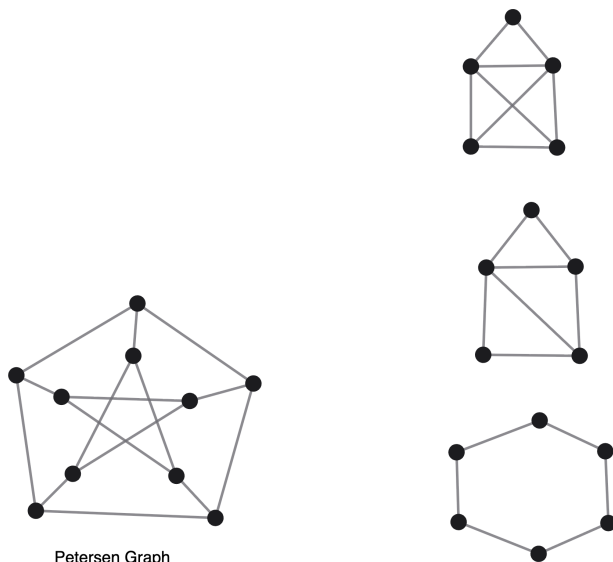


1. For a connected graph $G = (V, E)$ we define the diameter of G to be the maximum distance between any two vertices in V , where the distance between vertices is the minimum number of edges in a path connecting them.
 - (a) Find a formula for the diameter of the complete graph K_n for every n .
 - (b) Find a formula for the diameter of the hypercube graph H_n for every n .

2. Determine both the minimum and maximum number of leaves in a spanning tree for the following graphs



3. Prove or disprove that if $G = (V, E)$ is a connected graph with $|E| \geq |V|$ then G contains a cycle
4. Prove that if a graph $G = (V, E)$ is a tree, then for any edge $e \in E$, we have that $G' = (V, E \setminus \{e\})$ is disconnected.